Project-1: Distribution Comparison

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This mini project aims to compare the distribution of random exponential numbers against sample means of random exponential numbers and display the latter is approximately normal.

0. Import

```
suppressPackageStartupMessages({
  library(tidyverse)
  library(e1071)
  library(car)
})
```

1. Distribution of Random Exponential Numbers

Code below will randomly create 1 sample (40 observations) of exponential numbers, take their mean and variance, compare them with the theoretical parameters and graph their sample distribution.

```
set.seed(42)

rate=0.2
n <- 40

nums <- rexp(n, rate=rate)
mean_nums <- mean(nums)
mean_theoretical <- 1/rate
cat("Mean of random exponential numbers:", mean_nums, "\n")
```

```
## Mean of random exponential numbers: 4.915756
```

```
cat("Theoretical mean:", mean_theoretical, "\n")
```

```
## Theoretical mean: 5
```

```
var_nums <- var(nums)
var_theoretical <- 1/rate^2
cat("Variance of random exponential numbers:", var_nums, "\n")
```

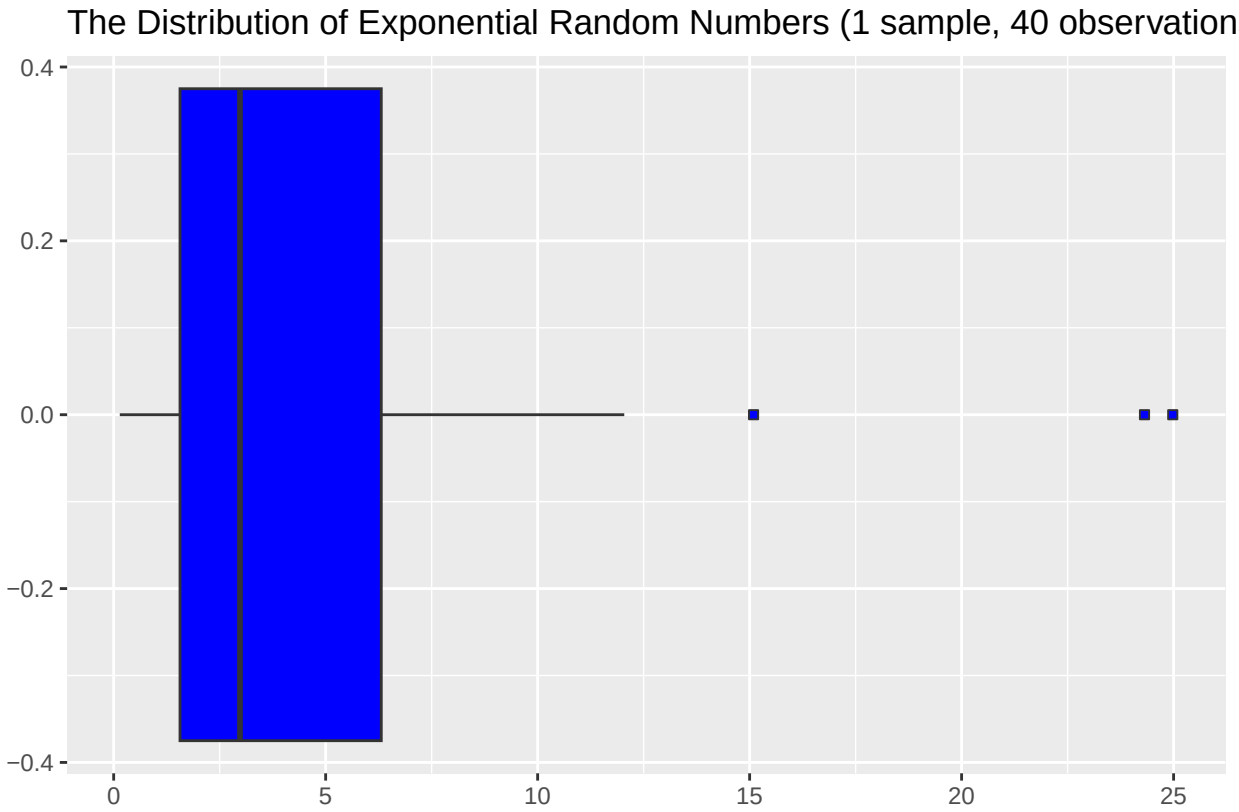
```
## Variance of random exponential numbers: 32.03207
```

```
cat("Theoretical variance:", var_theoretical)
```

```
## Theoretical variance: 25
```

```
as.data.frame(nums) |>
  ggplot(aes(x=nums)) +
  geom_boxplot(fill="blue", outlier.shape=22) +
```

```
  theme_grey() +
  labs(title="The Distribution of Exponential Random Numbers (1 sample, 40 observations)", x="")
```



2. Distribution of Sample Means of Random Exponential Numbers

Code below will randomly create 1000 samples (40 observations each) of exponential numbers, take sample means, take mean of sample means and graph "mean of sample means" distribution.

```
sample_size <- 1000

means <- apply(matrix(rexp(n=n*sample_size, rate=rate), sample_size), 1, mean)
mean_means <- mean(means)
cat("Mean of sample means of random exponential numbers:", mean_means, "\n")
```

```
## Mean of sample means of random exponential numbers: 4.987835
```

```
cat("Theoretical mean:", mean_theoretical, "\n")
```

```
## Theoretical mean: 5
```

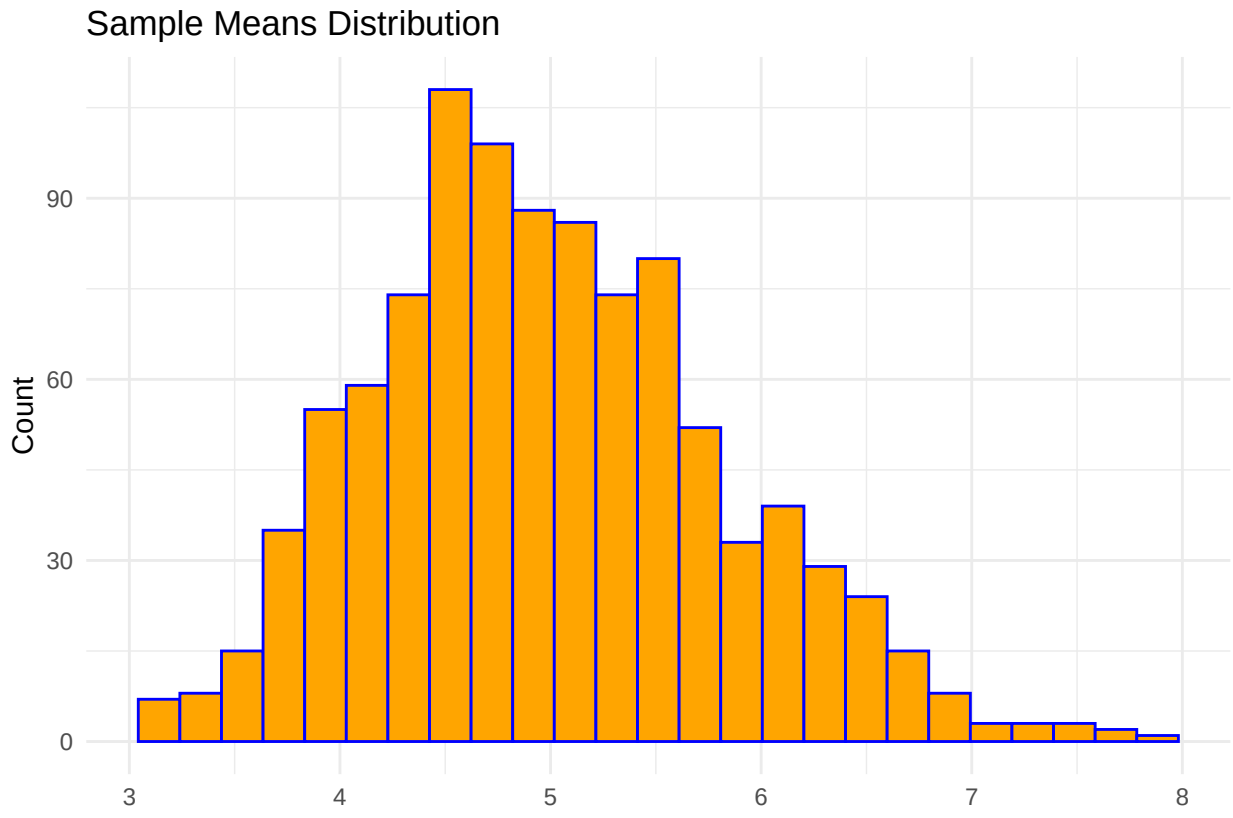
```
var_means <- var(means)
var_theoretical <- 1/rate^2/n
cat("Variance of random exponential numbers:", var_means, "\n")
```

```
## Variance of random exponential numbers: 0.6780786
```

```
cat("Theoretical variance:", var_theoretical)
```

```
## Theoretical variance: 0.625
```

```
as.data.frame(means) |>
  ggplot(aes(x=means)) +
  geom_histogram(bins=25, fill="orange", color="blue") +
  labs(title="Sample Means Distribution", x="", y="Count") +
  theme_minimal()
```



We have two results to declare:

1. Means of random exponential numbers and sample means of random exponential numbers are quite similar.
2. While the distribution of random exponential numbers is an exponential distribution as expected, the distribution of sample means of random exponential numbers is gaussian alike.

3. Normality Approximation of Sample Means Distribution

1. Skewness

   Code below will determine skewwness of the distribution of sample means

```
skew <- skewness(means)
cat("Skewness is: ", skew, "\n")
```

```
## Skewness is:  0.4217359
```

```
if(abs(skew)>0.5){
  cat("Considerable skewness!, Normality approximation is low.")
```

```
} else{
  cat("Slight skewness!, Normality approximation is high.")
}
```

```
## Slight skewness!, Normality approximation is high.
```

2. Kurtosis

   Code below will determine kurtosis of the distribution of sample means

```
kurt <- kurtosis(means)
cat("Kurtosis is: ", kurt, "\n")
```

```
## Kurtosis is:  -0.0214275
```

```
if(abs(skew)>1){
  cat("Considerable kurtosis!, Normality approximation is low.")
} else{
  cat("Slight kurtosis!, Normality approximation is high.")
}
```

```
## Slight kurtosis!, Normality approximation is high.
```

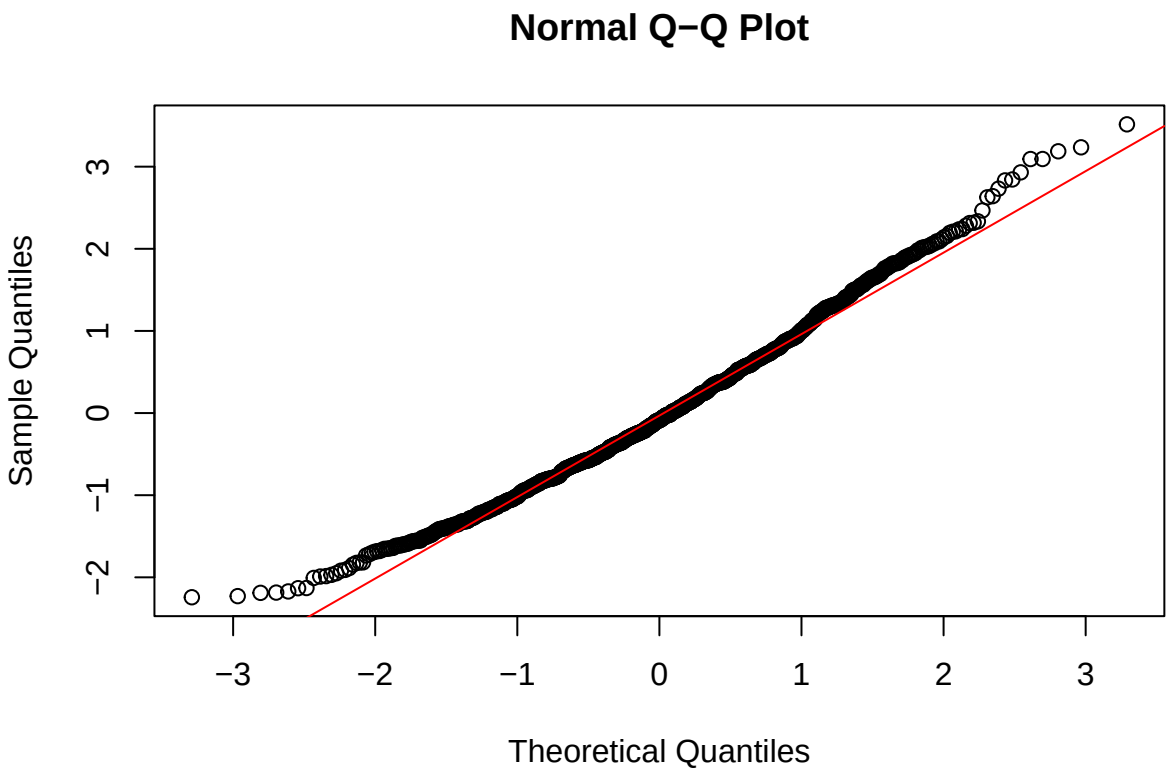
3. QQ Plot

   Code below will make a comparison between quantiles of the distribution of sample means against standard normal quantiles and graph them.

```
means_scaled <- scale(means)

qqnorm(means_scaled)

qqline(means_scaled, col = "red")
```



It is obvious that the sample means approximates normality considerably.